Sudhakar Alla

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Professional Summary

Senior Machine Learning Engineer who builds and ships production ML systems across autonomous robotics, industrial vision, and enterprise platforms. Owns the full model lifecycle end to end, from problem framing and data pipeline design through training, evaluation, deployment, and production monitoring. Brings a disciplined evaluation practice: defining what correct means, designing the metrics, and catching failure modes before they reach production. Led technical direction for AI/ML systems across six international markets, from architecture through stakeholder delivery.

Skills

Languages: Python, C++, SQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, CNNs, Vision Transformers / foundation models, feature embeddings, transfer learning, clustering, regression, model evaluation, ablation studies

Computer Vision: Object detection (YOLO, Faster R-CNN, SSD), segmentation, visual SLAM, structure from motion (SfM), multi-view geometry, camera calibration, 3D reconstruction, depth & pose estimation, coordinate frame transforms

Deployment & MLOps: PyTorch → ONNX → TensorRT, CUDA acceleration, embedded inference (Jetson), model serving, multithreaded batch inference, production monitoring, drift detection (PSI)

Data Engineering: PySpark, Databricks, Azure Data Factory, ETL pipelines, data ingestion

Tools & Infrastructure: OpenCV, NumPy, SciPy, Pandas, scikit-learn, scikit-image, Flask, Docker, APScheduler

Experience

- **Independent Machine Learning Engineer** — Remote, CA
  Self-employed – Client Projects — *January 2026 - Present*
  - **Scan-to-BIM Reconstruction**: Built a classical-geometry pipeline converting **225M+** point LiDAR scans of buildings into a structured 3D model: iterative RANSAC plane extraction, normal-based semantic classification, and instance clustering. Designed a per-class geometric self-consistency metric to define correctness without ground truth, surfacing confident misclassifications before downstream consumption.
  - **Road Defect Detection & Navigation (Repair Robotics)**: Built an embedded perception and navigation pipeline for autonomous road-defect detection from infrared depth camera imagery: DL-based segmentation for defect identification, classical CV post-processing for localization, and a full ROS navigation stack with georeferenced waypoint following. Real-time embedded inference via patch-based tiling and detection stitching.

- **Kimberly-Clark (via ICURO, then TechMahindra)** — Remote, CA
  AI Tech Lead — *April 2021 - February 2026*
  - **Forecasting & Promotion Optimization**: Sole author of a hierarchical forecasting pipeline feeding a promotion simulation-and-optimization engine for weekly planning across international markets.
    - Coverage: **100%** UK product coverage, **40%+** forecast accuracy improvement.
    - Performance: Reduced runtime **57%** (**72s → 31s**) via caching, scenario reuse, and grid-step tensor operations.
  - **Price Elasticity 2.0 (State Space / UCM)**: Developed and led UK production rollout of the second-generation elasticity estimation, controlling for confounders to isolate the causal effect of price on demand.
    - Subsequently deployed across **5 markets**; saving **$185K/year** in the UK and low seven-figure annual savings globally.
  - **Drift Monitoring & Production Ownership**: Led and mentored a rotating team of ML engineers and data scientists across regional deliveries, mentored engineers from junior to senior, and instituted PSI-based drift monitoring with automated retraining and stop triggers.

- **ICURO** — Santa Clara, CA
  Founding Autonomous Systems Engineer — *August 2018 - March 2023*
  - **Multi-Sensor Perception & Localization for Teleoperation**: Architected the perception and localization pipeline for an autonomous robotics teleoperation platform, from camera calibration and visual SLAM through 3D reconstruction and operator-facing path planning. Built the core back-projection system converting 2D operator inputs into 3D waypoints through sensor-to-world coordinate frame transforms and projection geometry, enabling reliable teleoperation across remote deployments.
    - Fused fiducial pose with feature-based VSLAM/SfM in an EKF, with confidence-weighted covariance and graceful failover when fiducials drop out of view.

  - Built a probabilistic occupancy grid from sonar and 2D LiDAR via an inverse sensor model, consumed by the planner for obstacle-aware trajectory generation.
  - Impact: Adopted across the platform by hundreds of operators in **daily production use**.
- **Crack Detection (Drone Perception)**: Designed and deployed a hybrid CV and deep learning pipeline for road crack detection from drone imagery. Used DL-based segmentation to isolate road surfaces, then applied image processing techniques (connected component analysis, pixel-intensity tracing) for crack localization. Optimized the full pipeline with TensorRT + CUDA on Jetson Xavier, reducing inference latency by **70%** for real-time embedded operation.

- **Applied Materials** — Santa Clara, CA
  Machine Vision Engineer — *June 2019 - June 2020*
  - **Defect Detection & Clustering (SEM Image Analysis)**: Solved an automated defect clustering problem three prior engineering teams had failed to solve. Combined transfer learning with domain-specific features to train CNN-based feature embedding models for defect representation learning and clustering under limited labeled data; achieved **95%+** cluster quality per process engineer evaluation. Designed the evaluation metrics and validation workflows to measure robustness across varied image conditions, and delivered a backend API for UI integration. – *In daily use by process engineers; authored an internal IEEE-style technical paper.*
  - **ML Model Serving & Inference Platform**: Designed the inference serving architecture for production ML workflows, implementing a queue-based system with multithreaded batching, preprocessing, and TensorFlow Serving integration; results persisted to database for job state tracking *–Deployed to daily production use by process engineers.*

- **Mental Scholar** — Sacramento, CA
  Software Engineer — *Summer 2012 (Intern), June 2013 – January 2017*
  - Developed technical tooling and visual content for academic math education, contributing to published e-books and an iOS application.

Teaching & Leadership

- **University of California, Santa Cruz — Silicon Valley Extension** — Santa Clara, CA
  Lecturer, AISV 807: AI for Robotics — *2021 - 2025*
  - **Curriculum design & mentoring**: Designed and delivered an applied AI-for-robotics course covering perception, SLAM fundamentals, and deployment pipelines; incorporated modern vision architectures (DINOv2, YOLO) for real-time detection projects and mentored student capstones including a vision-based robotic arm grasping project.

Education

- **California State University at Sacramento (CSUS)** — Sacramento, CA
  Bachelor of Science in Computer Engineering — *2016 - 2018*
  Tau Beta Pi Engineering Honor Society
- **American River College** — Sacramento, CA
  Transfer to CSUS & Certificates in Art New Media (2D/3D Animation) — *2013 - 2016*
  Transferred from Cal Poly, San Luis Obispo in 2013 to work with Mental Scholar